

Searching for Type Ia Supernovae in ZTF Deep Fields: Development of an Open-Source Framework

FLORIAN DEVENDER-DAUGE¹

¹*Ecole Universitaire de Physique et d'Ingénierie, UCA*

Abstract

Type Ia supernovae (SNe Ia) are crucial tools in observational cosmology, serving as standardizable candles due to their consistent peak luminosity after empirical corrections. Their use enabled the discovery of the Universe's accelerated expansion. However, their detection is still limited by instrumental sensitivity, survey depth, and limitations in data analysis pipelines. In this study, we explore whether stacking publicly available science images from the Zwicky Transient Facility to create deep field images (ZTF) can improve the detection rate of faint SNe Ia.

We first develop a semi-automated processing pipeline that aligns, calibrates, and cleans ZTF images quadrant by quadrant, before generating reference images using pixel-wise medians. We then perform temporal stacking of three consecutive exposures to enhance the signal-to-noise ratio and reveal transients below the standard detection threshold. The data are processed with a custom Python framework, **AstroTools**, which integrates image alignment, sky background subtraction, normalization via zero-point correction, and parallel processing using **ThreadPoolExecutor**.

A theoretical model based on photometric depth gain and source count scaling laws suggests that, under ideal observing conditions, up to three times more Type Ia supernovae could be detected compared to those listed in the *ZTF-COSMO-DR2 SNe Ia* catalog. This optimistic increase, however, is tempered by real-world limitations such as seeing degradation, sky background fluctuations, and instrumental artifacts, which reduce the expected gain. Nonetheless, these results highlight the strong potential of image stacking to uncover overlooked SNe Ia in archival data, and support the development of a scalable, open-source framework for community-driven transient discovery.

Keywords: ZTF, Deep Field, Sn Ia, Cosmology, Standard Candle, Database, Image Staking, Index

Contents

1. The Zwicky Transient Facility (ZTF)	4
1.1. Technical aspect of the telescope	4
1.2. Detection Pipeline	4
2. Type Ia supernovae and their role in cosmology	5
2.1. Why study type Ia supernovae?	5
2.2. Distance, magnitude, and Phillips relation	5
2.3. Cosmological implications and current debates	6
2.4. Challenges and biases in SNeIa standardization	7
3. Restructuring public data ZTF	8
3.1. Aim of our study	8
3.2. Original data structure	8
3.3. Supernovae Search Strategy	9
3.4. From Octets to Square Degrees of Sky	9
3.5. Data Collection and Filtering Methodology	10
4. Image processing and deep-field access	11
4.1. Pre-processing scientific images	11
4.2. Reference image construction	13
4.3. Stacking images for supernova detection	13
4.4. Hardware acceleration and scaling-up	13
5. Impact of image stacking on the number of detectable type Ia supernovae	14
5.1. Theoretical Impact	14
5.2. Estimated impact of difficulties encountered	15
6. Toward an Open-Source Framework: Future Directions	16
6.1. First Light Curve: Case Study of ZTF17aadlxmv	16
6.2. Deployment of an Open-Source Library	17
7. Futures prospects	18
7.1. An Open Science Approach to Transient Detection	18
7.2. Next targets	18

INTRODUCTION

As part of the end of our master’s degree in fundamental physics, I carried out this supervised project with my fellow student *Quentin ARVOIS* on the detection of type Ia supernovae in the deep field of ZTF. Strictly speaking, ZTF does not have a deep field, and we assume that stacking images will enable us to highlight phenomena that have not previously been observed. Our aim here is to achieve sufficiently efficient processing to be able to detect new supernovae in the same data as that used to produce the *ZTF-COSMO-DR2 SNe Ia* catalog (Rigault 2025).

Type Ia supernovae (SNe Ia) are critical tools in modern astrophysics, particularly for measuring extragalactic distances. Thanks to their relative uniformity in peak luminosity, they serve as standardizable candles and have played a pivotal role in unveiling the accelerated expansion of the Universe, one of the major breakthroughs in observational cosmology (Riess 1998; Perlmutter 1999). Beyond their cosmological relevance, detecting and cataloging SNe Ia is also essential for understanding stellar evolution in the time-domain sky.

Despite advances in instrumentation and sky surveys, the detection of SNe Ia remains limited by several factors: small sensitivity to faint events, high background levels, weather-dependent seeing conditions (a score calculated for each raw image, a function of the quality of the sky), and the limitations of real-time alert pipelines. One particularly underexploited resource is the growing archive of publicly available survey data, especially from large-scale time-domain projects like the Zwicky Transient Facility (ZTF).

ZTF, operating since 2018, produces millions of images across wide areas of the sky using a mosaic of 16 CCD detectors, each one separated in 4 quadrants. Its real-time alert system is optimized for fast transient discovery, but may overlook dimmer or marginal events due to thresholding or prioritization constraints. The archival science images, however, retain the full photometric information and represent a treasure trove for reanalysis provided that appropriate tools and methods are available.

In this context, the present work investigates whether image stacking (i.e., the combination of multiple exposures of the same field) can improve the detectability of SNe Ia in ZTF deep fields. By increasing the effective signal-to-noise ratio, stacking allows us to push the limiting magnitude deeper and potentially uncover transients that would otherwise remain undetected.

To do so, we designed and implemented a custom image processing pipeline capable of aligning, normalizing, cleaning, and combining ZTF images. A central component of this pipeline is the Python class we developed **AstroTools**, to manage quadrant-level processing, perform sky background correction using Zero-point calibration, and create reference images using pixel-wise medians. Empirical filters are applied to account for variations in seeing and image quality. The pipeline is appropriate for larger-scale processing in upcoming projects since it is reproducible, modular, and scalable.

1. THE ZWICKY TRANSIENT FACILITY (ZTF)

1.1. Technical aspect of the telescope

The *Zwicky Transient Facility (ZTF)* is a wide-field time-domain survey operating since 2018 at the Palomar Observatory. It officially began scientific operations in March 2018 at the Palomar Observatory in California. The project builds on the legacy of previous programs such as the *Palomar Transient Factory (PTF)* and its successor, the *Intermediate Palomar Transient Factory (iPTF)* (Law 2009; Rau 2009).

Mounted on the 48-inch Samuel Oschin telescope, ZTF is equipped with a 576-megapixel camera that covers a field of view of 47 square degrees per exposure, the largest of any optical survey camera in operation during its time (Bellm 2019). Its camera is composed of a mosaic of 16 CCD sensors, each with a resolution of 6144×6160 pixels. It operates in three photometric bands (g , r , and i) allowing multicolor tracking of observed objects. The detailed layout of the CCDs and their internal division into readout quadrants is shown in Figure 1.

ZTF scans the *entire visible northern sky night*, alternating between g and r bands (and occasionally i). This rapid cadence enables near-continuous monitoring of transient events¹, including supernovae, with a typical limiting magnitude of about 20.5 in 30-second exposures (Masci 2019).

Thanks to these features, ZTF has already led to the discovery of numerous remarkable celestial phenomena: type Ia supernovae detected sometimes only a few hours after their explo-

¹ A transient event is a short-lived celestial event, such as a supernova, stellar flare or neutron star collision. These events are often detected by their variation in luminosity or other observable characteristics over relatively short periods of time.

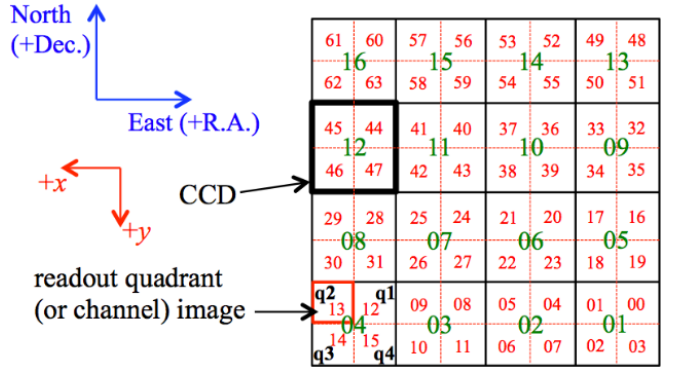


Figure 1. Layout of the ZTF camera focal plane, showing the 16 CCDs arranged in a 4×4 grid. Each CCD is divided into four readout quadrants, which are the base units for image calibration and processing (Masci 2020).

sion, unusual variable stars, solar system objects such as near-Earth asteroids. These discoveries highlight the program’s effectiveness in capturing fleeting events in their earliest stages (Golkhou 2022; Dekany 2020).

For SNe Ia, this means that ZTF can detect them *shortly after explosion*, often within 1 to 3 days, which is critical for studying the early-time light curve behavior and probing explosion physics (Olling 2015)

To identify transients, ZTF employs *image differencing*: a newly acquired image is subtracted from a reference image of the same sky region, allowing detection of new sources. This is done using the *Hotpants* software, a robust difference imaging tool (Becker 2015).

1.2. Detection Pipeline

Each night, ZTF generates *tens of thousands of transient alerts*, which are streamed in real-time to brokers such as Lasair and ALeRCE that filter and distribute them to the scientific community for follow-up (Patterson 2019). This sys-

tem allows for rapid discovery and classification of SNe Ia.

A key component is the *Bright Transient Survey (BTS)*, which targets nearly all transients brighter than $m \sim 18.5$ for spectroscopic classification. BTS ensures a *nearly complete sample* of low-redshift SNe Ia, serving as cosmological anchors and machine-learning training sets for LSST (Perley 2020).

ZTF’s cadence allows detection of *early-time flux excesses*, as seen in *SN 2018oh*, interpreted as possible ejecta-companion interaction (Shappee 2019). Simulations suggest that ZTF’s cadence permits *early detection* (within 3 days post-explosion) in 10 to 15% of cases (Yao 2019).

Ultimately, ZTF provides a *critical statistical foundation* for supernova cosmology. Its large, well-characterized sample of nearby SNe Ia bridges the gap between local calibrators and high- z events from future surveys such as *LSST* (Ivezić et al. 2019).

2. TYPE IA SUPERNOVAE AND THEIR ROLE IN COSMOLOGY

2.1. Why study type Ia supernovae?

Type Ia supernovae (SNe Ia) result from the thermonuclear explosion of a white dwarf in a binary system. Two main explosion scenarios are proposed: the *single-degenerate* scenario, in which the white dwarf accretes matter from a companion star until it reaches the Chandrasekhar limit ($\sim 1.4 M_\odot$), and the *double-degenerate* scenario, involving the merger of two white dwarfs. The exact mechanism remains debated, but the photometric properties of SNe Ia are remarkably homogeneous (Hillebrandt 2000).

This homogeneity allows their use as *standard candles*, that is, as distance indicators. The

Phillips relation (Phillips 1993) shows that the maximum luminosity of a SNe Ia is correlated with the width of its light curve: brighter supernovae decay more slowly. The use of SNe Ia as cosmological probes led to the discovery of the acceleration of the expansion of the Universe (Riess 1998; Perlmutter 1999), a major breakthrough that led to the introduction of the concept of dark energy.

Their immense brightness ($M_B \approx -19.3$) make SNe Ia observable out to redshifts $z \sim 1$ by the largest telescopes and far beyond the reach of variable stars.

2.2. Distance, magnitude, and Phillips relation

The apparent magnitude m of a supernova and its absolute magnitude M are related by the formula:

$$\mu = m - M = 5 \log_{10}(D_L/10 \text{ pc}), \quad (1)$$

where μ is the distance modulus and D_L is the luminosity distance in parsecs. This formula, based on the relationship between luminosity and distance, is one of the fundamental tools for measuring cosmological distances in astronomy.

The apparent magnitude m of an object is defined as the measure of its brightness observed from Earth, while the absolute magnitude M represents the intrinsic brightness of the object at a distance of 10 parsecs. A lower magnitude indicates a brighter object. This logarithmic scale is known as the *Pogson relation*, formulated as:

$$m - m_0 = -2.5 \log_{10} \left(\frac{I}{I_0} \right), \quad (2)$$

where I is the intensity of the object and I_0 is the intensity of a standard reference (Pogson 1856).

Even before the introduction of the Phillips relation, a more generalized empirical formula-

tion known as the Tripp relation (Tripp 1998) was commonly used. It introduces a linear correction to the absolute magnitude of Type Ia supernovae, accounting for both the stretch of the light curve and the observed color. It is usually expressed as:

$$\mu = m_B - M_B + \alpha(s - 1) - \beta C, \quad (3)$$

where m_B is the apparent peak magnitude, M_B the standardized absolute magnitude, s the stretch factor of the light curve, C the color (often defined as $B - V$), and α , β are empirical coefficients calibrated from observational data.

The Phillips relation can be viewed as a special case of this formulation, considering only the stretch term. The Tripp relation has thus become a foundational tool in modern cosmological analyses involving Type Ia supernovae.

The Phillips relation adjusts M based on the light curve decay of supernovae. Specifically, this relation shows that the maximum luminosity of a SNe Ia is correlated with the rate of decay of its light curve, allowing for the standardization of supernovae brightness. The Phillips relation can be expressed as follows:

$$M_B = M_B^0 + \alpha \times (s - 1), \quad (4)$$

where s is the time stretch factor of the light curve, and α is an empirical coefficient (~ 1.5) (Phillips 1993). Thus, the combination of the Pogson relation and the Phillips relation allows type Ia supernovae to serve as *standard candles*, reliable and precise distance indicators in the universe.

2.3. Cosmological implications and current debates

In observational cosmology, Type Ia supernovae (SNe Ia) have become essential tools, particularly for the discovery of the Universe’s accelerated expansion, which paved the way for

the development of the Λ CDM model. The way our universe expands is a function of the value of the Hubble Constant, measuring how fast a far object moves away from us (Leibundgut, Bruno 2018). Discovering more SN Ia will help cosmologists to be more accurate about this constant, in the middle of a tension between them.

This tension results from a large and ongoing discrepancy between the value of the Hubble constant H_0 measured directly in the local universe and that extrapolated from early-universe observations.

On one hand, measurements of the cosmic microwave background (CMB)—particularly those from the *Planck* satellite assuming the Λ CDM model—yield $H_0 \approx 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Collaboration 2020).

On the other hand, local measurements based on the distance ladder method using Cepheid variables to calibrate Type Ia supernovae suggest a significantly higher value of $H_0 \approx 73.0 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Riess 2019, 2022).

This discrepancy (more than 5σ), illustrated in Figure 2, has persisted despite increased precision on both sides. It is now one of the most pressing puzzles in observational cosmology, and could point to unknown systematics or to new physics beyond the standard Λ CDM model. Among the proposed solutions are:

- **Early dark energy:** a brief phase of accelerated expansion before recombination, which could alter the sound horizon at decoupling and reconcile the two H_0 estimates (Poulin et al. 2019).
- **Neutrino physics modifications,** including extra relativistic species (parametrized as ΔN_{eff}) or a time-varying neutrino mass (Archidiacono et al. 2013).

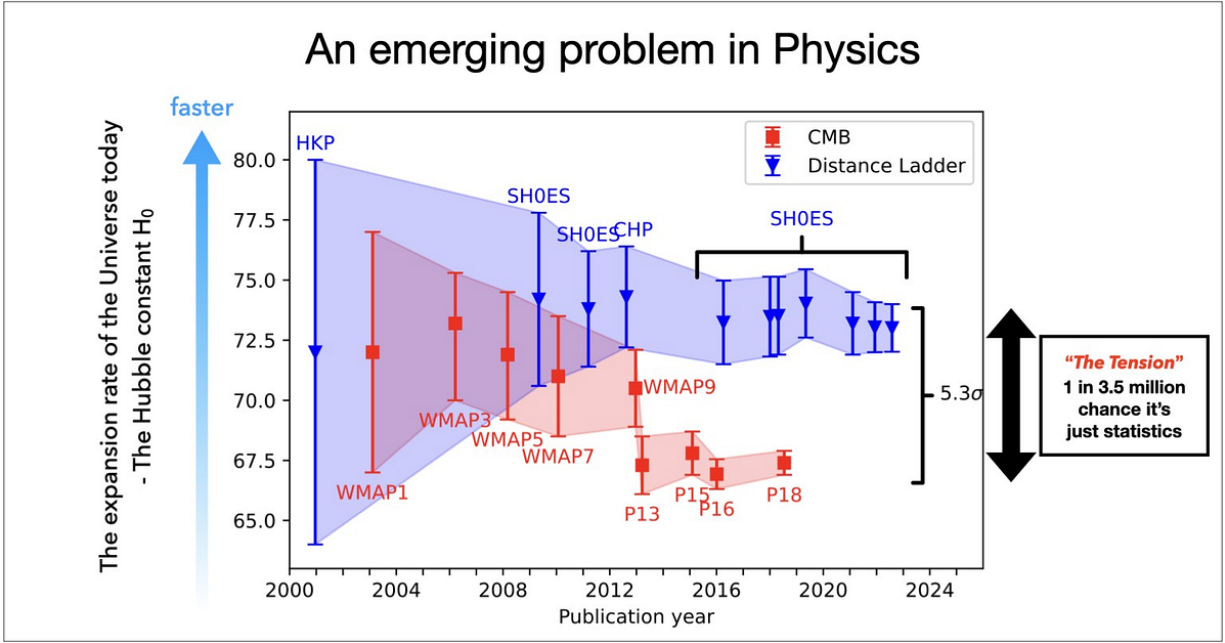


Figure 2. Comparison of the Hubble constant H_0 derived from early-universe observations (Planck, blue) and late-universe observations (SH0ES, red) (Association of Universities for Research in Astronomy (AURA) 2023)

- **Modified gravity or dark sector interactions**, which would change the Universe’s expansion history or the growth of structure (Di Valentino et al. 2020).
- **Systematic biases** in SNe Ia or Cepheid calibration, although the robustness of local measurements has improved substantially with *Gaia* parallaxes and refined statistical techniques (Riess 2022).

SNe Ia therefore play a dual role: not only are they central to local H_0 determination, but their standardizable nature across cosmic time also makes them key probes for testing the redshift evolution of cosmological parameters.

Upcoming surveys such as those from the Vera C. Rubin Observatory (LSST), *Euclid*, and the *Nancy Grace Roman Space Telescope* are expected to greatly increase the number and precision of high-redshift SNe Ia, providing essential data to resolve the tension or to confirm the need for new physics beyond Λ CDM.

2.4. Challenges and biases in SNeIa standardization

Despite their utility, SNe Ia are not perfectly standard candles. Several known biases remain sources of systematic uncertainty:

- **Host galaxy dependence:** More massive or older host galaxies tend to host slightly brighter SNe Ia. This effect, known as the “mass step,” typically amounts to ~ 0.1 mag and is often corrected for in cosmological analyses based on host stellar mass or local star formation properties (Sullivan et al. 2010).
- **Redshift evolution:** The progenitor populations of SNe Ia may evolve with redshift, due to changing metallicity or star formation history in galaxies. While current studies such as Pantheon+ do not find significant evolution for $z < 1$, this remains an area of active research (Brout et al. 2022).

- **Intrinsic dispersion and calibration:**

Even after empirical corrections (e.g., Tripp relation), a residual scatter of ~ 0.1 – 0.2 mag persists. Moreover, absolute magnitude calibration depends on the anchoring method—such as Cepheids, the Tip of the Red Giant Branch (TRGB), or maser-based distances—which directly impacts the value of H_0 (Riess 2022).

3. RESTRUCTURING PUBLIC DATA ZTF

3.1. *Aim of our study*

The proposed approach consists of downloading public ZTF science images and co-adding them to increase the instrument’s limiting magnitude. Stacking multiple images should allow the detection of fainter objects, including supernovae that were not bright enough to trigger a real-time alert. These supernovae could then be characterized using their light curves, although the resulting uncertainties would be larger than for unstacked data due to signal-to-noise propagation. With this method, we expect to uncover significantly more SNe Ia than those listed in the official ZTF supernova catalogue for the 2018-2020 period.

3.2. *Original data structure*

One of the main challenges of this project lies in accessing and exploiting the public data provided by the Zwicky Transient Facility (ZTF). These data, hosted on the IRSA (Infrared Science Archive) servers, are stored in strictly chronological order (Masci 2019).

While this organization may seem straightforward at first, it becomes a real obstacle when one aims to retrieve all observations of a specific region of the sky across an extended time

period, a necessary condition for plotting a usable light curve for a Type Ia supernova.

In fact, to correctly track the photometric evolution of such an event, observations spanning **at least 60 to 100 days** are typically required. However, because the IRSA archives are sorted by date rather than by sky region, it is very difficult to access, for each day, the exact area of the sky corresponding to the supernova’s location. This limitation greatly complicates any effort to reconstruct the temporal evolution of a given source.

In practical terms, each observation made by ZTF corresponds to a 30-second exposure. After each exposure, the telescope slews to a new field, generating a sequence of snapshots of the sky (Masci 2020). Each of these snapshots is associated with a timestamp, referred to as a *fractime*, which encodes the observation time in millionths of a day (i.e., it is derived from the HH:MM:SS time of the exposure).

To determine which part of the sky was imaged at a given fractime, one must access the specific URL hosted by IRSA that contains metadata for that observation. Over the entire period studied—spanning **1044 nights** from March 3, 2018, to January 10, 2021, and covering nearly **544,000 individual exposures** (Rigault 2025) this task becomes monumental.

The first naive approach involved downloading the HTML page corresponding to each observation day, scanning it to extract all listed fractimes, and checking for each one whether it pointed to the correct sky region (among nearly 900 fields) and the correct filter (among the three used: g , r , and i). Assuming a typical distribution of fractimes across all fields and filters, this approach led to an average of **2700**

pages to scan per day, resulting in a bandwidth usage of roughly **270 MB per day**.

This method, although functional, was clearly inefficient. It required enormous computational resources, and was painfully slow to implement. Extracting a single month’s worth of useful data could take several days of processing, which was incompatible with the large-scale goals of the project.

This realization led to the development of a more robust and scalable system. Rather than evaluating each HTML page in real time, the new method relies on an offline parsing and structuring of the data: first downloading raw file headers and organizing them into a local database, which can then be queried efficiently based on both temporal and spatial criteria. This shift from a manual, exploratory approach to an automated and optimized pipeline represents a key turning point in the project. It laid the foundation for building a truly open and reproducible framework for future studies based on ZTF public data.

3.3. *Supernovae Search Strategy*

A few days before the start of the work presented here, the ZTF collaboration released its initial catalog of SNIa, comprising 3268 objects observed between March 3, 2018, and January 10, 2021 (Rigault 2025), spanning 1044 days. Assuming that every night the entire sky observable by ZTF was surveyed without interruption due to weather conditions, covering an area of 33,660 square degrees (Dekany 2020), we can thus calculate a maximum sky coverage equal to:

$$33,660 \times 1,044 = 35,141,040 \quad (5)$$

This maximum surface of 35,141,040 square degrees was scanned at most over 1044 days. That being said, from a practical standpoint,

the telescope could not conduct data acquisition campaigns every night. I therefore developed a program to determine how many nights were actually recorded. The result: only 828 nights provided usable data, corresponding to a duty cycle of 79.3%.

3.4. *From Octets to Square Degrees of Sky*

To assess the effective sky coverage of ZTF over the observed period as accurately as possible, we implemented an empirical estimation method based on the structure of the public data files. We need to be accurate because it is our way of estimating the number of new Supernovae we could find. These files, provided in `.html` formats, follow a highly regular format which we exploited to convert raw file sizes—expressed in bytes—into astronomical sky area, expressed in square degrees.

Each observation day on the IRSA archive corresponds to a single HTML page listing all exposures (or *fractimes*) for that night. These pages follow a rigid structure: a fixed-length header of 315 characters, a footer of 22 characters, and a middle section containing all the `fractime` entries. Each `fractime` line, which encodes one observation, is exactly 39 bytes long. Since each `fractime` corresponds to a unique telescope exposure covering 47 square degrees of the sky, this allows a direct proportionality between the number of fractimes and the total covered area.

We thus deduce the number of `fractime` entries per file by subtracting the fixed-length header and footer (a total of 337 bytes) from the total size of the file, and dividing by the length of a single `fractime` entry (39 bytes). Multiplying this number by 47 then yields the area covered by that file in square degrees. The formula becomes:

$$S = \sum_{i=1}^n (Z_n - 337) \times \frac{47}{39} \quad (6)$$

Where:

- S is the total sky area covered (in square degrees),
- Z_n is the size of the n -th file in bytes,
- 337 accounts for the sum of the fixed header (315 bytes) and footer (22 bytes),
- 39 is the exact size (in bytes) of a single `fractime` entry,
- 47 is the known sky coverage per ZTF exposure, in square degrees.

However, summing over each individual file is computationally unnecessary if one knows the total number of files N and their cumulative size Z . We can thus derive a simplified expression:

$$S = (Z - 337 \times N) \times \frac{47}{39} \quad (7)$$

This compact formulation allows us to compute sky coverage efficiently for large datasets. For instance, using empirical data from 828 observation files with a total byte count of $Z = 21,808,011$ bytes, we compute:

$$S = \frac{(21,808,011 - 337 \times 828) \times 47}{39} \\ \approx 25,945,175 \text{ square degrees.}$$

This value represents the cumulative sky area scanned by ZTF during the 828 nights of usable data. Although this is an overestimate, it serves as a cornerstone for estimating detection rates. For example, knowing that the catalog of Rigault et al. (Rigault 2025) lists 3268 type Ia supernovae detected over the same period, we derive the average density of detections per unit area and per night as:

$$\frac{3268}{25,945,175} \approx 1.26 \times 10^{-4} \text{ SNIa deg}^{-2} \text{ night}^{-1}.$$

The typical anisotropic sky coverage of ZTF, shaped by its survey strategy and galactic plane avoidance, is illustrated in Figure 3.

3.5. Data Collection and Filtering Methodology

Faced with the limitations of the initial strategy, it quickly became necessary to design a more structured and scalable method for collecting and organizing the public ZTF data. The main objective was to shift from a reactive and time-consuming process to a proactive system capable of selecting only the relevant observations, field by field, according to well-defined scientific criteria (Masci 2019).

The chosen solution was to download all available index pages from the IRSA servers approximately **544,000 HTML files**, locally converted into `.txt`, representing a total volume of about **50 GB of data**. From these files, a script was developed to systematically extract all `fractime` entries and build a single `.csv` file indexed by sky region. This centralized file enables rapid identification of all dated observations for a given area of the sky, without the need to repeatedly query the IRSA website (Masci 2020).

Unlike the initial approach, this new method relies on a **one-time massive pre-processing** operation, which drastically reduces processing time during the actual targeted download phases. Although this version does not yet include advanced filtering by photometric band (the g , r , and i bands are still mixed in the results), it already leads to a major reduction in unnecessary requests. In practice, it is now possible to **download several months' worth of data in less than one hour**, compared to several days using the previous method.

Each observation listed in the `.csv` file is associated with its observation date, sky region

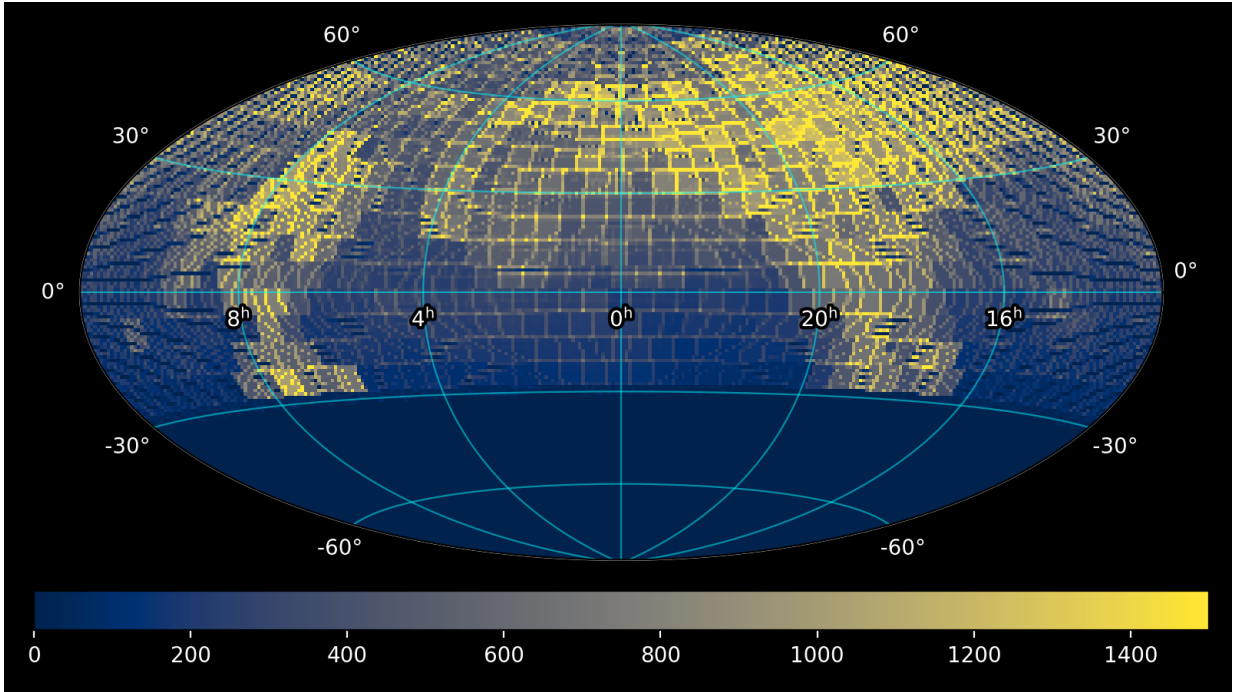


Figure 3. Sky coverage map from the Zwicky Transient Facility (ZTF) Data Release 7 (IPAC - Infrared Processing and Analysis Center 2023). Brighter areas indicate regions with higher observational cadence and deeper exposure, revealing the tiling geometry and prioritization of survey fields across the northern sky.

(ZTF field), `fractime`, and its corresponding download link. This local pre-indexing system provides a solid foundation for automating the entire data collection pipeline while allowing great flexibility in the selection of regions of interest.

For instance, a specific download can now be triggered with a single command, as shown below:

```
download_images_from_csv(
    csv_path="fichiers_par_field.csv",
    target_field="0625",
    filter_band="zr",
    ccd="06",
    quadrant="1",
    start_date="20200406",
    end_date="20210803")
```

Put simply, this approach allows a shift from inefficient manual exploration to a large-scale, automated management system, fully aligned

with a high-throughput analysis strategy. It also serves as a progressive base for future improvements, such as filter-specific selection or integration with other transient event catalogs.

4. IMAGE PROCESSING AND DEEP-FIELD ACCESS

In order to improve the depth of detection of Type Ia supernovae in the ZTF's public data, an image processing pipeline has been designed to pre-process the science images, generate stable reference images and perform temporal stacking. This processing relies heavily on the work of *Quentin ARVOIS*, whose expertise has made it possible to structure the entire processing flow in an efficient, modular and reproducible way.

4.1. Pre-processing scientific images

The scientific images downloaded from the ZTF public archive are multi-extension FITS files. Each file corresponds to a CCD sensor,

Custom Science Image Processing Pipeline by Quentin Arvois

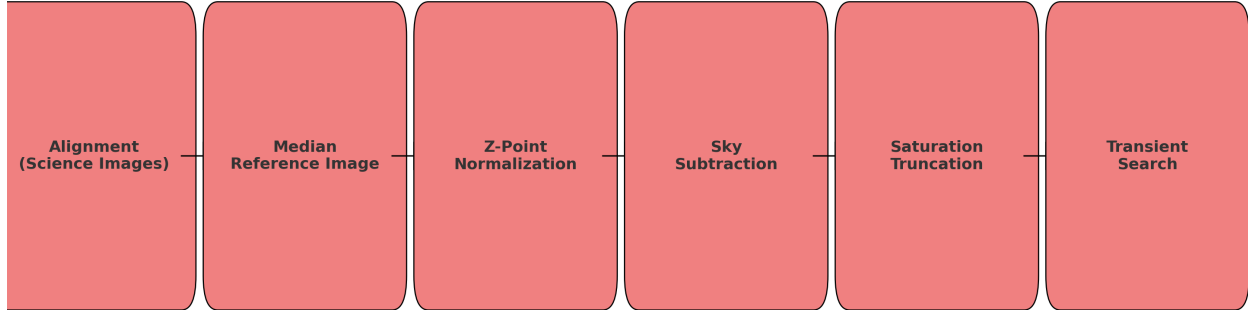


Figure 4. Simplified diagram of the custom processing pipeline applied to ZTF science images. Each raw image first undergoes photometric normalization using the Z-point value from the FITS headers, followed by quadrant-by-quadrant astrometric alignment. The aligned images are then cropped to retain only overlapping pixel regions, and combined using a pixel-wise median to produce a robust reference image. Sky background is estimated and subtracted to enhance contrast, and outlier pixels (e.g., saturated or anomalous values) are masked.

itself divided into four readout channels, called quadrants. Analyses are carried out quadrant by quadrant to ensure consistent processing.

The first processing operation carried out by *Quentin ARVOIS* consists of precisely aligning the scientific images with each other, based on the WCS (World Coordinate System which associates each pixel of an image to a sky position in equatorial (ra,dec) coordinates information). Several operations are then applied to make the images comparable:

- **Photometric correction** : the intensities are normalised using the value *Z-Point* present in the FITS headers, in order to guarantee a homogenous photometric scale between exposures.
- **Troncature of saturated pixels**: values above the saturation threshold are eliminated to avoid bias when subtracting images.
- **Background subtraction**: a median value is estimated for each quadrant and removed to improve contrast with astrophysical objects.

These actions were first carried out as separate functions that were manually executed for every FITS file list. Despite working for initial testing, this approach soon showed itself to be inefficient for processing on a big scale. Consequently, a more resilient architecture took its place.

This is the **AstroTools** class.

All of the processing functions were consolidated into a Python class named **AstroTools** in order to guarantee the pipeline's clarity, reusability, and reproducibility. This class is the main interface for all image analysis operations (see fig 4):

- It is initialized with a list of paths to FITS files corresponding to a given ZTF quadrant. An optional identifier is used to label the output in a structured way.
- The class methods handle alignment, normalisation, reference image generation, subtraction, photometry and statistical analysis of seeing.

Among its notable features, the seeing analysis method produces summary statistics and a histogram with Gaussian adjustment. This diagnostic is used to assess the quality of input

images before stacking, to avoid degraded images introducing artefacts in subsequent stages.

4.2. Reference image construction

Reference images play a central role in the detection of transients, since they serve as the basis for image subtraction. The aim is to represent the sky with no transient events, a stable background and as few artefacts as possible.

The procedure used is as follows:

1. **Selection** of the images corresponding to the same CCD quadrant over a given period.
2. **Astrometric alignment** via the WCS, then **spatial cropping** to retain only the pixels present in all the images (intersection zone).
3. **Calculation of the median** for each pixel, in order to obtain an image that is representative of the celestial background, without being polluted by outliers or point transients.

The choice of the median rather than the mean proved to be decisive: during the first tests, using the mean introduced visible anomalies during subtraction. The median, on the other hand, effectively eliminates cosmic artefacts, hot pixels and non-persistent transient events.

The result is a reliable, stable reference image that can be used to detect new objects using differential subtraction.

4.3. Stacking images for supernova detection

Once the images have been calibrated and the references constructed, the next step is to stack

several successive images of the same field to improve sensitivity to weaker sources.

In this work, we opted for stacking in groups of three images. This choice is based on several considerations:

- offers a signal-to-noise gain of the order of $\sqrt{3}$ (i.e. about +0.6 magnitudes).
- maintains a sufficient time cadence to re-construct a usable light curve.
- maintains a reasonable volume of data compatible with rapid analysis.

The images are first aligned and then cut in the same way as references, and the combination is typically carried out by average (or median in certain special cases). Unlike reference images, the aim of these ‘stacked’ images is not to smooth transients, but on the contrary to reveal them with a better SNR (Masci 2019).

They are then subtracted from the corresponding reference image, producing a residual map in which supernovae clearly appear as excess point sources. This method thus makes it possible to detect events that would have escaped the ZTF alert pipeline because of their low individual luminosity.

4.4. Hardware acceleration and scaling-up

The increase in the volume of data processed means that we need to optimize performance. The ZTF’s observation campaigns produce several hundred thousand FITS images, making sequential processing unsuitable for large-scale analysis.

To address this issue, part of the pipeline has been optimised using the `ThreadPoolExecutor` library, which comes from the standard `concurrent.futures` module in Python. This

library enables independent tasks to be parallelized efficiently, making full use of the processor’s available cores.

The following steps have been successfully parallelized:

- alignment and cropping of individual images;
- the calculation of the median for the generation of reference images ;
- quadrant-by-quadrant image subtraction operations;
- conversion and writing of intermediate files.

Each task is dynamically allocated to a thread in the pool, saving significant time without manual management overhead. On a standard quad-core machine, this optimisation allows the total processing time for a given field to be divided by a factor of 3 to 4.

This move to a multi-threaded architecture is an essential milestone in the systematic processing of hundreds of ZTF fields. It positions this project within a semi-automated production logic, compatible with large-scale scientific exploitation ([Python Software Foundation 2023](#)).

Coupled with the modular structuring of the **AstroTools** class, this optimisation not only enables fast execution but also great flexibility in resource allocation, facilitating future adaptation towards distributed computing or the use of processing clusters.

5. IMPACT OF IMAGE STACKING ON THE NUMBER OF DETECTABLE TYPE IA SUPERNOVAE

5.1. *Theoretical Impact*

This section presents a quantitative estimate of the increase in the number of detectable type Ia supernovae (SNe Ia) made possible by stacking multiple images of the same region of the sky. The objective is to assess whether this technique can significantly improve the scientific return from existing public data ([North 2014](#)).

We consider here a single CCD quadrant covering approximately one quarter of a ZTF field, i.e., 11.75 deg^2 , observed over the entire available period of 357 nights (as determined during the data acquisition phase). Using the average detection rate previously established ($1.26 \times 10^{-4} \text{ SNIa deg}^{-2} \text{ night}^{-1}$, see Section 3.4), the expected number of detectable SNe Ia without any image stacking is:

$$\begin{aligned} \langle N_{\text{SNIa}} \rangle &= 1.26 \times 10^{-4} \times 11.75 \times 357 \\ &\approx 0.53 \text{ SNIa} \end{aligned}$$

This corresponds to fewer than one event over the entire period. The total data volume associated with these 357 nights (assuming 16 CCD quadrants per night and $\sim 38 \text{ MB}$ per FITS file) is approximately:

$$0.038 \text{ GB} \times 16 \times 357 \approx 217 \text{ GB}$$

The goal of image stacking is to increase the signal-to-noise ratio (SNR), allowing the detection of fainter transient objects that remain below the detection threshold in individual exposures. When stacking N images, the signal grows linearly with N , while the noise increases only as $\sqrt{N}$ ([Schroeder 1999](#)). This yields an improvement in limiting magnitude Δm expressed as:

$$\Delta m = 1.25 \log_{10}(N) \quad (8)$$

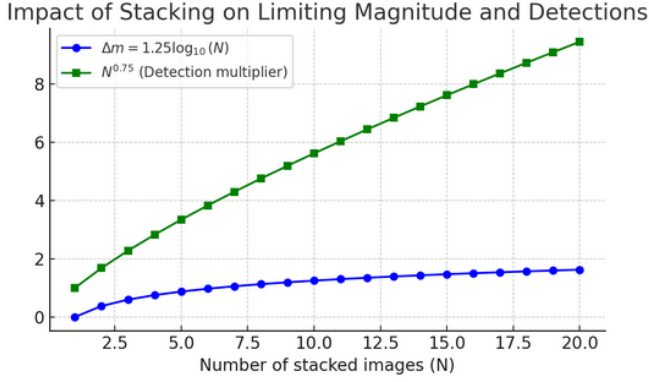


Figure 5. Expected gain in limiting magnitude (Δm) and relative increase in the number of detectable objects ($N^{0.75}$) as a function of the number of stacked images N .

In deep photometric surveys, the cumulative number of detectable sources typically follows a power-law distribution:

$$N(m) \propto 10^{\alpha m} \quad (9)$$

with $\alpha \approx 0.6$ (North 2014) and m the apparent magnitude. The stacking-induced gain in detectable objects is then:

$$N_{\text{stacked}} = N_{\text{initial}} \times N^{1.25\alpha} \approx N_{\text{initial}} \times N^{0.75} \quad (10)$$

For instance, stacking $N = 4$ images results in an expected increase in the detection rate by a factor of $4^{0.75} \approx 2.82$. Applying this to our initial estimate:

$$\langle N_{\text{SNIa}} \rangle_{\text{stacked}} = 0.53 \times 2.82 \approx 1.49 \text{ SNIa}$$

Figure 5 summarizes this trend, showing the expected gain in limiting magnitude and detection rate as a function of the stacking depth N .

It is important to note, however, that this estimate relies on idealized assumptions and does not account for several real-world factors:

- partial overlap or variation in sky coverage between nights,

- changes in seeing conditions and sky background brightness,
- inaccuracies in photometric normalization and astrometric alignment.

Nonetheless, even under conservative assumptions, image stacking emerges as a promising and scalable method to improve the detection efficiency for faint transients, especially those missed by the default ZTF alert pipeline. This strategy is particularly well-suited for exploiting deep public survey data, enabling discoveries without requiring additional telescope time.

5.2. Estimated impact of difficulties encountered

Our approach has the potential to uncover a greater number of type Ia supernovae than those reported in existing ZTF catalogs, by reaching fainter magnitudes and revisiting archival data that were previously discarded or overlooked. However, a critical perspective is necessary when interpreting these results.

First, because of the decreased signal-to-noise ratio close to the detection threshold, some of the discovered candidates can have lower confidence levels. To verify these occurrences, other procedures like PSF photometry or human validation might be needed. Second, there is still a problem that hasn't been solved at scale in this framework yet: we need to develop a software to automatically create light curves of transients and skip the artifacts.

Furthermore, a significant fraction of the public ZTF data proves to be poorly usable due to unfavorable observational conditions. High background levels or degraded seeing can strongly affect the quality of stacked images, especially when faint sources are involved. Our method incorporates filtering based on seeing

statistics and background estimates to mitigate these effects, but some degradation is inevitable.

Consequently, the optimistic factor of 2.82 in the enhanced detection yield (obtained under optimal circumstances) is probably exaggerated. Until we haven’t yet found another supernovae, we cannot find a better estimation of this factor. This should still a significant gain, though, and it makes the effort to use automated techniques to scale up the analysis worthwhile.

6. TOWARD AN OPEN-SOURCE FRAMEWORK: FUTURE DIRECTIONS

6.1. *First Light Curve: Case Study of ZTF17aadlxxmv*

To validate the end-to-end pipeline developed in this project, we applied it to the supernova ZTF17aadlxxmv, a confirmed type Ia event detected in early 2020. Using science images obtained in the ZTF-r filter, we extracted a light curve by stacking triplets of aligned exposures and applying pixel-level photometry on the resulting difference images.

Figure 6 shows the resulting flux evolution over time. The light curve exhibits the expected photometric signature of a typical type Ia supernova. A sharp brightness increase is observed in early January 2020, reaching its peak around January 30 consistent with the predicted time of maximum light in thermonuclear explosion models.

Following this maximum, the brightness decreases smoothly. Notably, a distinct secondary feature emerges around mid-February: a “shoulder” or secondary peak, characteristic of type Ia supernovae in red and near-infrared bands. This phenomenon is attributed to the recombination of iron-group elements in the ejecta, and its presence further supports

the classification of ZTF17aadlxxmv as a type Ia event.

In late March, the flux continues to decline, eventually settling into a low plateau phase in April. The increased scatter in this phase is likely due to lower signal-to-noise ratios and sparser sampling, yet the global temporal structure remains intact.

COMPARE WITH ZTF CATALOG DATA

To evaluate the accuracy of our subtraction and photometry pipeline, we compared our extracted light curve with the official ZTF catalog data for the same object. Both datasets are shown in Figure 7. Our measurements (blue curve) and the catalog fluxes (orange curve) were each smoothed using a 12th-order polynomial fit to better highlight their temporal structure. Despite a clear offset in absolute flux largely due to differing photometric zero-points (27.5 for our pipeline vs. 30.0 in the ZTF catalog) the overall shapes of the two light curves are remarkably similar. Both display:

- a rapid flux rise in early January,
- a well-defined maximum around January 28–31,
- a secondary maximum in mid-February,
- and a smooth decline into a low-plateau regime by April.

Attempts to directly convert our instrumental fluxes to the ZTF scale led to inconsistencies and unphysical values, likely due to differences in photometric conventions (e.g., flux units, area normalization, or filter calibration) not explicitly documented in the FITS headers. However, these scaling differences do not affect the morphological agreement, which confirms the reliability of our pipeline for temporal reconstruction (Rigault 2025).

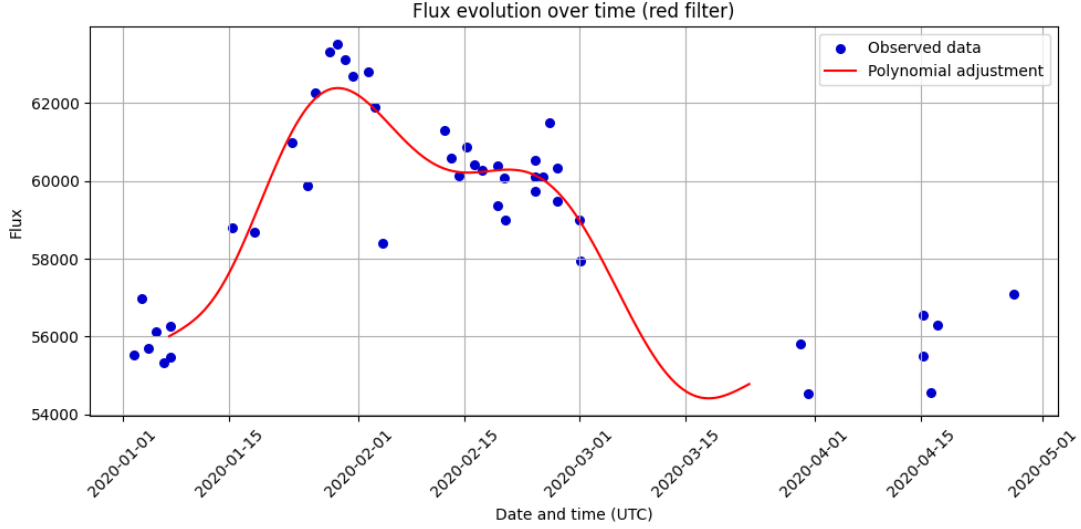


Figure 6. Flux light curve of ZTF17aadlxxmv extracted using the AstroTools pipeline (ZTF-r filter). The temporal evolution shows a primary maximum at the end of January 2020, followed by a characteristic shoulder in mid-February, consistent with the behavior of type Ia supernovae.

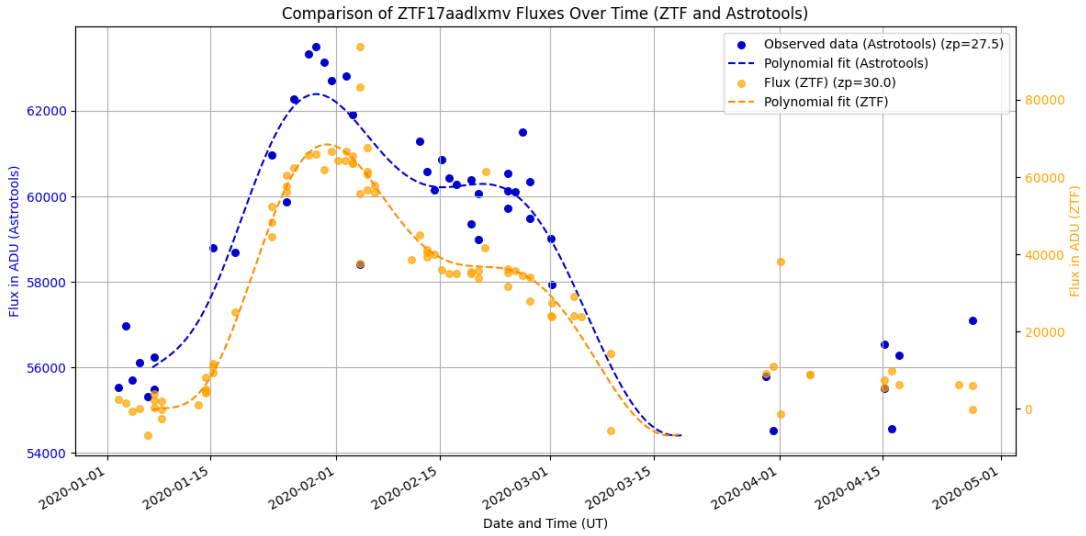


Figure 7. Comparison between the flux light curve extracted using the AstroTools pipeline (blue) and the ZTF catalog data (orange) for ZTF17aadlxxmv. Both datasets were fitted with a 12th-order polynomial to enhance visibility. While the absolute flux values differ, the temporal evolution and main features align well, validating the accuracy of the custom pipeline.

6.2. Deployment of an Open-Source Library

To facilitate reproducibility, modularity, and community collaboration, all tools developed in this project have been integrated into a custom Python library, tentatively named **ZTF-Tools**. This library encapsulates all stages of the data processing pipeline, including:

- automated parsing and filtering of ZTF FITS metadata;
- quadrant-level alignment and sky normalization;
- construction of median reference images;

- difference imaging and candidate identification;
- light curve extraction and visualization.

The `AstroTools` class serves as the central interface, providing both flexibility for interactive analysis and batch-mode compatibility for large-scale processing. The entire codebase is structured to be portable, readable, and easily extensible. The goal is to release this toolset under an open-source license, with clear documentation and tutorials, to foster adoption by other researchers and citizen scientists.

7. FUTURES PROSPECTS

7.1. *An Open Science Approach to Transient Detection*

In the long term, this approach could scale into a community-driven platform for faint transient discovery. As datasets grow in size and complexity, decentralized processing becomes increasingly attractive. An open-source stacking pipeline like the one developed here could serve as a lightweight alternative to proprietary pipelines, extending the scientific reach of archival data. While the initial results are promising, several improvements are required to fully operationalize this framework:

- **Photometric calibration:** At present, light curves are generated in instrumental flux units. A conversion to physical magnitudes via reference stars or ZTF calibration catalogs is needed for proper astrophysical interpretation.
- **Artifact rejection:** Despite median filtering, residual cosmic rays, CCD defects, and variable stars can still appear in subtracted images. More advanced filtering (e.g., PSF fitting, machine learning) could improve purity.
- **Seeing normalization:** A tighter integration of seeing distribution analysis into the stacking process could reduce variability in image quality and improve SNR homogeneity.
- **Scalability:** Although multi-threaded processing via `ThreadPoolExecutor` accelerates the pipeline, deployment on HPC clusters or cloud environments could enable massive field-scale processing.

7.2. *Next targets*

In the month following the submission of this report, the project will continue with a strong focus on open science. The major purpose will be to adapt the current pipeline into a fully described, installable open-source Python library, enabling other academics or sophisticated hobbyists to use it to explore the ZTF archives.

Simultaneously, a Type Ia supernova automated detection module will be created that incorporates preprocessing, picture subtraction, transient identification, and a confidence scoring system. In order to enable community-driven contributions to the finding of hitherto undiscovered supernovae in public datasets, the objective is to develop a completely autonomous framework that can automatically highlight the most likely candidates after scanning an entire area.

CONCLUSION

Assessing whether picture stacking techniques could enhance the identification of type Ia supernovae in archival data from the Zwicky Transient Facility was the aim of this master’s research project. This project provided me, as a master’s student, the opportunity to deeply explore the complexities of data processing, instrumentation constraints, and the practical application of an entire analysis pipeline, all while conducting a scientific inquiry. We demonstrated that this approach is both technically feasible and scientifically valuable by developing a modular, semi-automated program capable of aligning science images, adjusting photometric backgrounds, and building reference and stacked images with pixel-level precision.

Early theoretical models suggested that stacking as few as three images could enhance detection efficiency by up to 2.8 times under ideal observational conditions. While modest, this improvement is significant, particularly when considering that it relies solely on the reprocessing of publicly available data, without any need for additional telescope time. This highlights the power of leveraging existing resources to enhance our understanding of the universe, particularly in cosmology, where precision measurements of supernovae distances and luminosities play a critical role in refining our models of dark energy and the accelerated expansion of the universe.

The pipeline was used to reconstruct the light curve of a known type Ia supernova from the ZTF dataset, serving as a proof of concept. While this was not a new discovery, it validated the entire toolchain and demonstrated that it could consistently recover key photometric properties, including the primary peak and secondary maximum, with minimal human intervention. This step was pivotal in establishing the scientific credibility of the method.

From a student’s perspective, this project provided a comprehensive introduction to scientific research, from conceptualization to computational implementation and critical analysis. It offers a solid, transparent, and repeatable foundation for future research, even though it does not represent a groundbreaking shift in transient astronomy. Future advancements, such as confidence scoring for transient candidates, refined photometric calibration, and the extension to large-scale field surveys, will be greatly facilitated by the combination of open-source development, flexible data management, and multithreaded processing.

Ultimately, this research has yielded a valuable tool that can be understood, modified, and expanded by others. The methodology developed here contributes to democratizing the discovery of transients by providing easy access to underutilized data and accelerating the process from raw images to light curves that can be used for scientific analysis. This opens up opportunities for both amateur and academic researchers to engage in the search for faint, obscure astronomical phenomena, furthering our understanding of the universe.

Finally, ZTF is often regarded as a precursor to the forthcoming LSST (Legacy Survey of Space and Time), operated by the Vera C. Rubin Observatory. Both projects share a wide-field observational philosophy and a strong emphasis on the automated processing of vast data sets, which will be essential for advancing cosmological studies, such as the precise measurement of dark energy’s role in the universe’s accelerated expansion ([Collaboration 2019](#)).

ACKNOWLEDGEMENTS

I would like to express my deep gratitude to my supervisor, *Philippe ROSNET*, and his colleague *Marie AUBERT* for their support, availability, and invaluable advice throughout this work. Their expertise and kindness were crucial to the success of this project.

Many thanks to my colleague and friend *Quentin ARVOIS*, for his work on aligning the images and creating the references, and his unfailing support, without which a large part of this project would not have been possible.

I'd also like to thank my family and friends, especially my wife *Natasha* and my father *Yannick*, for their moral support, patience, and unfailing encouragement during these months of work and these years of studies.

Finally, I would like to thank the team at the Zwicky Transient Facility and the IRSA archive for making their data publicly available, without which this project would not have been possible.

REFERENCES

- Maria Archidiacono, Elena Giusarma, Alessandro Melchiorri, and Olga Mena. Neutrino and dark radiation properties in light of recent cmb observations. *Phys. Rev. D*, 87:103519, May 2013.
<https://doi.org/10.1103/PhysRevD.87.103519>.
 URL <https://link.aps.org/doi/10.1103/PhysRevD.87.103519>.
- Association of Universities for Research in Astronomy (AURA). Our mysterious universe still evades cosmological understanding, March 2023. URL <https://www.aura-astronomy.org/blog/2023/03/06/>.
 Accessed: June 5, 2025.
- Andrew C. Becker. Hotpants: High order transform of psf and template subtraction. <http://ascl.net/1504.004>, 2015. Astrophysics Source Code Library.
- S. R. Graham M. J. al. Bellm, Eric C. Kulkarni. The zwicky transient facility: System overview, performance, and first results. *Publications of the Astronomical Society of the Pacific*, 131(995):018002, 2019.
- D. Brout, D. Scolnic, B. Popovic, and et al. The pantheon+ analysis: Cosmological constraints. *The Astrophysical Journal*, 938(2):110, 2022.
<https://doi.org/10.3847/1538-4357/ac8e04>.
- LSST Collaboration. The vera c. rubin observatory legacy survey of space and time, 2019. Available at <https://www.lsst.org/>.
- Planck Collaboration. Planck 2018 results. vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.
<https://doi.org/10.1051/0004-6361/201833910>.
- R. others Dekany. The zwicky transient facility: Observing system. *PASP*, 132(1009):038001, 2020.
<https://doi.org/10.1088/1538-3873/ab4ca2>.
- Eleonora Di Valentino, Alessandro Melchiorri, Olga Mena, and Sunny Vagnozzi. Nonminimal dark sector physics and cosmological tensions. *Physical Review D*, 101(6), March 2020. ISSN 2470-0029.
<https://doi.org/10.1103/physrevd.101.063502>.
 URL <http://dx.doi.org/10.1103/PhysRevD.101.063502>.
- V. Z. others Golkhou. Ztf early observations of supernovae: Identifying explosions within hours of explosion. *ApJ*, 926(1):38, 2022.
<https://doi.org/10.3847/1538-4357/ac3b34>.
- J. C. Hillebrandt, W. Niemeyer. Type ia supernova explosion models. *Annual Review of Astronomy and Astrophysics*, 38:191–230, 2000.
<https://doi.org/10.1146/annurev.astro.38.1.191>.
- IPAC - Infrared Processing and Analysis Center. Ztf releases seventh data set, November 2023. URL <https://www.ipac.caltech.edu/news/dr7>.
 Accessed: June 5, 2025.
- Ž. Ivezić, S. M. Kahn, J. A. Tyson, and et al. Lsst: From science drivers to reference design and anticipated data products. *The Astrophysical Journal*, 873(2):111, 2019.
- N. M. others Law. The palomar transient factory: System overview, performance, and first results. *PASP*, 121:1395, 2009.
<https://doi.org/10.1086/648598>.
- Dhawan, Suhail Jha, Saurabh W. Leibundgut, Bruno. Measuring the hubble constant with type ia supernovae as near-infrared standard candles. *AA*, 609:A72, 2018.
<https://doi.org/10.1051/0004-6361/201731501>.
 URL <https://doi.org/10.1051/0004-6361/201731501>.
- F. J. others Masci. The zwicky transient facility: Data processing, products, and archive. *PASP*, 131(995):018003, 2019.
<https://doi.org/10.1088/1538-3873/aae8ac>.
- Frank J. et al. Masci. Ztf science data system explanatory supplement v5.0, 2020. Available at https://www.ipac.caltech.edu/doi/ztf/ztf_explanatory_supplement.pdf.
- N. North, G. James. *Observing Variable Stars, Novae and Supernovae*. Cambridge University Press, 2014. ISBN 9781107636125. URL <https://books.google.fr/books?id=IzoDBAAQBAJ>.
- R. Shaya E. J. al. Olling, R. P. Mushotzky. Shortly after the explosion: Early light curves of type ia supernovae from the kepler space telescope. *Nature*, 521:332–335, 2015.
- F. J. Laher R. al. Patterson, M. T. Masci. Real-time alert distribution for the zwicky transient facility. *Publications of the Astronomical Society of the Pacific*, 131(995):018001, 2019.

- C. Yan L. al. Perley, D. A. Fremling. The zwicky transient facility bright transient survey. *The Astrophysical Journal*, 904(1):35, 2020.
- S. others Perlmutter. Measurements of ω and λ from 42 high-redshift supernovae. *ApJ*, 517: 565–586, 1999. <https://doi.org/10.1086/307221>.
- M. M. Phillips. The absolute magnitudes of type ia supernovae. *ApJ Letters*, 413:L105–L108, 1993. <https://doi.org/10.1086/186970>.
- Norman Pogson. Magnitudes of thirty-six of the minor planets for the first day of each month of the year 1857. *Monthly Notices of the Royal Astronomical Society*, 17(1):12–15, 11 1856. ISSN 0035-8711. <https://doi.org/10.1093/mnras/17.1.12>.
- V. Poulin, T. L. Smith, T. Karwal, and M. Kamionkowski. Early dark energy can resolve the hubble tension. *Physical Review Letters*, 122(22):221301, 2019. <https://doi.org/10.1103/PhysRevLett.122.221301>.
- Python Software Foundation. concurrent.futures — launching parallel tasks, 2023. Available at <https://docs.python.org/3/library/concurrent.futures.html>.
- A. others Rau. Exploring the optical transient sky with the palomar transient factory. *PASP*, 121:1334, 2009. <https://doi.org/10.1086/605911>.
- A. G. others Riess. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *AJ*, 116:1009–1038, 1998. <https://doi.org/10.1086/300499>.
- Adam G. al. Riess. Large magellanic cloud cepheid standards provide a 1% foundation for the determination of the hubble constant. *The Astrophysical Journal*, 876(1):85, 2019. <https://doi.org/10.3847/1538-4357/ab1422>.
- Adam G. al. Riess. A comprehensive measurement of the local value of the hubble constant with 1 km/s/mpc uncertainty from the hubble space telescope and the sh0es team. *The Astrophysical Journal Letters*, 934(1):L7, 2022. <https://doi.org/10.3847/2041-8213/ac5c5b>.
- Mickael al. Rigault. Zr2: The ztf supernovae ia reference catalog (2018–2021), 2025. Submitted. Data available at <https://github.com/ZwickyTransientFacility/ztfcosmo>.
- D.J. Schroeder. *Astronomical Optics*. Academic Press, 1999. ISBN 9780080499512. URL <https://books.google.fr/books?id=fu3650dUNPcC>.
- K. Z. Kochanek C. S. al. Shappee, B. J. Stanek. Early excess emission in the type ia supernova 2018oh. *The Astrophysical Journal*, 870(1):13, 2019.
- M. Sullivan, A. Conley, D. A. Howell, and et al. The dependence of type ia supernovae on their host galaxies. *Monthly Notices of the Royal Astronomical Society*, 406(2):782–802, 2010. <https://doi.org/10.1111/j.1365-2966.2010.16731.x>.
- R. Tripp. A two-parameter luminosity correction for type ia supernovae. *Astronomy and Astrophysics*, 331:815–820, 1998.
- A. A. Kulkarni S. R. al. Yao, Y. Miller. Ztf early observations of type ia supernovae reveal diverse light-curve features. *The Astrophysical Journal*, 886(2):152, 2019.